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B.Tech in Artificial Intelligence & Data Science
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Career Objective

Forward-Deployed AI Engineer with 2+ years of production experience architecting, evaluating, and operating enterprise LLM and RAG systems embedded on-site with large listed conglomerates (Adani Group). Proven track record delivering 10+ production microservices under the unified central enterprise gateway, with regulatory RAG pipelines operating autonomously in production for 12+ months. Specialized in multi-backend LLM failover, FinOps token optimization, custom PostgreSQL observability, deterministic verification guardrails, and automated daily C-suite reporting.

Experience

• AI Engineer (Forward-Deployed) **September 2025 – Present**
CognitBotz Solutions | *Client: Adani Green Energy & Adani Group* *Ahmedabad, Gujarat*

Platform 1: Aegis — Enterprise Compliance & Legal Intelligence Suite (6 Applications)

Enterprise compliance automation suite in 24/7 production for 12+ months across Corporate Legal and Secretarial teams.

- **Regulatory RAG Pipelines (SEBI, RBI, BSE):** Architected 3 production RAG pipelines ingesting daily exchange and central bank gazettes; built dynamic chunkers (single-turn for notices, hierarchical map-reduce for 200-page master directions), PyMuPDF OCR fallback, and multi-backend LLM orchestration (Azure OpenAI GPT-4.1-mini, Azure AI Llama-4-Maverick, self-hosted Phi-3 on on-prem Tesla T4 GPU) cutting regulatory noise by **71%** and saving **250+ FTE hours/quarter**.
- **FinOps Token Economics & On-Prem Routing:** Routed batch summarization for **50+ group entities** to self-hosted on-prem models combined with prompt caching, slashing recurring cloud LLM API expenses by **60%+** while maintaining sub-second inference.
- **Alphanumeric Regulatory ID Search:** Eliminated BPE tokenizer subword fragmentation on statutory circular IDs through deterministic regex normalization and hybrid lexical boosting, guaranteeing 100% rank-1 circular retrieval.
- **Insider Trading & Secretarial AI with HITL:** Engineered Databricks-backed Insider Trading surveillance pipelines, automated Director Disclosures, and an AI Board Minutes generation service featuring a **Human-in-the-Loop (HITL)** review drawer with page-level evidence citation prior to executive sign-off.
- **LLM-as-Judge Evaluation & Group ARB Clearance:** Built automated evaluation harness measuring RAGAS-aligned dimensions (Context Precision, Faithfulness, Answer Relevance, <1% hallucination); authored enterprise architecture specifications and presented directly to the **Group Architecture Review Board (ARB)**, securing enterprise security and architecture clearance.

Platform 2: Equity Dashboard — Capital Markets Intelligence Suite

Centralized equity intelligence portal across 7 listed entities integrating market RAG, research analytics, and investor CRM.

- **Shareholding Pattern (SHP) Intelligence & AI Analyst:** Multi-tenant analytics portal tracking **3.9M+ investor records**; engineered pre-LLM deterministic entity resolver (matching dates to settlement weeks and names to master lists), 4-tier confidence gate, and BU-scoped query planner achieving p50 ≈ **3ms** query latency; automated daily headless PDF briefings to Business Unit CEOs.
- **Sell-Side Broker Intelligence & Research RAG:** Extracted forward estimates (Revenue, EBITDA, PAT, EPS, Target Price) from broker PDFs via Azure AI Search (Hybrid BM25 + Cohere Embed v4.0 + Semantic Reranker); built deterministic 8-point numerical verification engine (quote matching, unit normalization, cross-statement sanity checks) with dual-LLM failover.
- **Equity Analytics RAG Platform:** Built conversational RAG system with zero-hallucination guarantees over BSE/NSE market and valuation data; engineered 3-tier embeddings across 10 tables evaluated on a 39-question golden benchmark.
- **Institutional IR Console:** Integrated investor relations meeting tracker and engagement analytics with role-based access control.

Platform 3: Debt Markets Central Enterprise Infrastructure

High-availability reverse proxy infrastructure orchestrating 10+ microservices alongside debt capital markets backends.

- **Debt Markets Backend Engineering:** Developed async FastAPI backend services and PostgreSQL data schemas for Mutual Fund debt exposure mapping, Pension Fund NPS tracking, and Debt Capital Markets counterparty meeting management (MCM).
- **Custom PostgreSQL Observability Gateway Architecture:** Configured central Nginx reverse proxy with SSL termination, HTTP/2, HSTS, and SSE streaming across 10 microservices; engineered a custom in-house AI observability platform tracking distributed execution traces, latency percentiles, and token metrics natively in Azure PostgreSQL; integrated Azure AD SSO traversing corporate Zscaler and Fortinet firewalls, cutting latency from **8s to 500ms**.

• AI Engineer **November 2024 – September 2025**
CognitBotz Solutions *Hyderabad, Telangana*

- **Production AI Platforms (DocCapture AI & Intelligent AI):** Architected and deployed 2 live commercial platforms — **DocCapture AI** (multi-modal document intelligence and automated table/entity capture) and **Intelligent AI** (enterprise low-code platform for rapid agentic workflow orchestration).
- **LLM Engineering & Agent Deployment:** Developed and fine-tuned LLMs using LangChain and Agno with few-shot prompting; built async FastAPI microservices supporting high-throughput multi-modal web agents.
- **AI Guardrails & Responsible AI:** Enforced strict Pydantic output schemas, prompt-injection sanitization, confidence gating, and data-privacy safeguards.

• AI Developer Intern **September 2024 – December 2024**
Infosys Springboard *Remote*

- **Enterprise NLP Workflows:** Built NLP and text classification pipelines for automated document routing under Infosys's enterprise track.

Projects

Project 1: Granite Buyer Intelligence Platform — Resilient Task Queue & 3-Tier LLM Gateway *granite.rhinofoster.com*

Deployed B2B market intelligence platform with durable asynchronous task execution and multi-provider failover.

- **Durable PostgreSQL Task Queue:** Built crash-resilient asynchronous task queue with 180s leases and heartbeat recovery for automated multi-engine web discovery, qualification, and data extraction.
- **3-Tier LLM Failover Gateway:** Implemented dynamic failover architecture (OpenRouter → NVIDIA Build → Groq) and lexical search, guaranteeing 100% inference uptime and sub-second generation.
- **Evidence-First Validation:** Enforced SQL check constraints separating verified facts from AI inference, with 0–100 quality scoring and audit tracking.

Project 2: Sentinel (Ward) — AI Model Governance & Security Platform *intelligentagentworks.com*

Deployed auditable on-ramp for running open-source models on owned GPUs with zero LLM in the critical path.

- **Deterministic Pickle-Opcode Scanner:** Engineered custom AST and pickle-opcode scanner detecting unsafe deserialization exploits (RCE) prior to loading model weights into GPU memory, completely bypassing LLM overhead.
- **Model Integrity & Policy Pipeline:** Implemented SHA-256 cryptographic verification, automated Hugging Face license classification, and a multi-stage cyber/legal governance approval workflow backed by FastAPI and React.

Project 3: PerkAI — Async Multi-Agent Subsidy & Benefits Discovery Platform *Deployed on Azure VM*

High-concurrency pure-asyncio multi-agent platform discovering cloud credits, grants, and enterprise tax subsidies.

- **Autonomous Multi-Agent Pipeline:** Architected pure-Python `asyncio` agent orchestrator (QueryPlanner, SearchService with circuit breakers, Extraction, Rule-Matching with geo-gating, Valuation) streaming execution via SSE to a React 19 SPA with Edge-TTS voice synthesis.
- **Resilient AI Gateway:** Built unified gateway routing chat/JSON across Groq and OpenRouter with 2048-d embeddings, automated JSON schema repair, and strict verbatim-quote grounding guards.
- **3-Tier Enterprise Observability:** Instrumented distributed tracing via Langfuse (prompt versioning, token metrics), Prometheus custom metrics, and Grafana/Loki log aggregation correlated by `X-Request-ID`.

Technical Skills

Agentic AI & LLMs: Multi-Agent Systems, LangGraph, LangChain, Agno, Tool/Function-Calling, State Graphs, Supervisor-Worker Patterns, Prompt Engineering, Prompt Caching, FinOps Token Optimization, Frontier Models (OpenAI, Claude, DeepSeek R1, Gemini, Azure OpenAI, Llama-4, Phi-3, Groq LPU)

Advanced RAG & Search: Hybrid Search (Dense Vector + BM25/FTS), Cross-Encoder Reranking (FlashRank, BGE-Reranker, Cohere Rerank), Azure AI Search, Parent-Child Chunking, Dynamic Hierarchical Chunking, Vector Databases (pgvector on PostgreSQL, FAISS, ChromaDB, Weaviate), Cohere Embed v4.0 (1536-dim)

AI Evals & Observability: LLM Evals (RAGAS, DeepEval, G-Eval, LLM-as-Judge), Observability & Distributed Tracing (Custom PostgreSQL AI Observability, Langfuse, OpenTelemetry), Failure Mode Analysis, Guardrails (Pydantic Schema Validation, NeMo Guardrails, Confidence & Abstention Gating, HITL Workflows)

Backend, Data & Cloud: Python, FastAPI, Pydantic v2, PostgreSQL (Azure Database Flexible Server), Databricks, Redis, Asyncio, React, TypeScript, Vite, Nginx (Reverse Proxy / Load Balancing), Docker Compose, Playwright, Azure (OpenAI, AI Foundry, AD SSO / OAuth2, Key Vault, VM), Tesla T4 Air-Gapped Serving, Group ARB Clearance

Awards & Certifications

• **Performance of the Year Award – GreenRatna Initiative** (2025–2026) **Adani Green Energy Limited**

Conferred by the Chief Digital Officer (CDO, AGEL) and recognized across the GreenRatna initiative for high-impact production delivery of enterprise AI compliance and capital markets platforms; appreciated by Group Chairman Shri Gautam Adani.

• **Certifications:** Anthropic Claude Prompt Engineering Certification | CCSK v4.1 Foundation (Cloud Security) | Cloud Computing Fundamentals | AICTE-Cisco Cyber Security | Google Cloud Bootcamp (GeeksforGeeks)

Education

• **Bachelor of Technology in Artificial Intelligence and Data Science** **2021 – 2025**

Terna Engineering College, Osmanabad (Dharashiv), Maharashtra *First Class with Distinction*